

TOPOLOGICAL GAMES AND BAIRE SPACES

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These notes study *topological games*: infinite games in which two players alternately choose open sets, and in which topological properties of a space are characterized by which player has a winning strategy. The two main results are that a space is a Baire space exactly when the first player has no winning strategy in the Banach–Mazur game (Theorem 2.3), and that a metrizable space is completely metrizable exactly when the second player has a winning strategy in the strong Choquet game (Theorem 3.4). Put together, they give a game-theoretic proof of the Baire category theorem (Corollary 3.6).

0. BAIRE CATEGORY

Definition 0.1. Let X be a topological space, and let $E \subseteq X$. The closure of E will be denoted by $\overline{E}$, and the interior of E by E° .

- (1) The set E is said to be *nowhere dense* in X if $(\overline{E})^\circ = \emptyset$. Equivalently, E is nowhere dense if every non-empty open set $U \subseteq X$ contains a non-empty open set V such that $V \cap E = \emptyset$.
- (2) The set E is said to be of the *first category* in X (or *meager*) if E is the union of a countable family of nowhere dense sets. A set which is not of the first category is said to be of the *second category* (or *non-meager*).
- (3) A set $C \subseteq X$ is said to be *comeager* (or *residual*) if the complement $X \setminus C$ is of the first category.
- (4) The space X is called a *Baire space* if the intersection of every countable family of dense open subsets of X is dense in X .



Remark 0.2. The notions in Definition 0.1 classify subsets according to topological “size,” providing a qualitative analogue of measure theory.

- (1) A nowhere dense set is thin everywhere: every non-empty open set contains a non-empty open set that misses the nowhere dense set entirely. Every finite subset of $\mathbb{R}$ is nowhere dense, and so is the Cantor ternary set.
- (2) A set of the first category is topologically negligible, and a countable union of such sets is again of the first category, just as a countable union of Lebesgue-null sets is null. The rationals $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ form a set of the first category in $\mathbb{R}$. The analogy has limits: $\mathbb{R}$ is the union of a set of the first category and a set of Lebesgue measure zero [Oxt80, Chapter 1], so “small” in one sense can be “large” in the other.
- (3) A comeager set contains “most” points of the space; the elements of a comeager set are called the *generic* points of X . In $\mathbb{R}$, the set of irrational numbers is comeager.
- (4) By Theorem 0.3 below, X is a Baire space exactly when no non-empty open subset of X is negligible. This is a *local* condition, and this condition is stronger than asking that X itself be of the second category in X : the disjoint union $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\}) \subseteq \mathbb{R}^2$ is of the second category in itself, because the open subset $\mathbb{R} \times \{1\}$ of X is, but X is not a Baire space, because the open subset $\mathbb{Q} \times \{0\}$ of X is of the first category.

Theorem 0.3. *A topological space X is a Baire space if and only if no non-empty open subset of X is of the first category.*

Proof. Suppose X is a Baire space, and let $G \subseteq X$ be open and non-empty. If $G = \bigcup_{n=1}^{\infty} E_n$ with each E_n nowhere dense, then $V_n = X \setminus \overline{E_n}$ is open and dense in X . Hence $\bigcap_{n=1}^{\infty} V_n$ is dense in X , so $\bigcap_{n=1}^{\infty} V_n$ must intersect G . However,

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction. Thus G is of the second category.

Conversely, suppose no non-empty open subset of X is of the first category. Let $\{V_n\}$ be a sequence of dense open subsets of X , and let $W \subseteq X$ be open and non-empty. Write $E_n = X \setminus V_n$. Each E_n is closed, so $\overline{E_n} = E_n$, and E_n has empty interior because V_n is dense; thus each E_n is nowhere dense. If $W \cap \bigcap_{n=1}^{\infty} V_n = \emptyset$, then $W \subseteq \bigcup_{n=1}^{\infty} E_n$. This makes W a non-empty open set of the first category, contrary to our hypothesis. Hence $W \cap \bigcap_{n=1}^{\infty} V_n \neq \emptyset$, and the intersection is dense. 

1. INFINITE GAMES

Definition 1.1. Let A be a non-empty set. We denote by $A^{<\omega}$ the collection of all finite sequences $s = (a_0, \dots, a_{n-1})$ of elements of A . The length n of s is denoted by $|s|$; the empty sequence $\emptyset$ has length 0. If $t \in A^{<\omega}$ and $|t| \leq |s|$ with $t_i = s_i$ for $0 \leq i < |t|$, we write $t \subseteq s$. For $s \in A^{<\omega}$ and $a \in A$, we write $s \frown a = (s_0, \dots, s_{|s|-1}, a)$ for the extension of s by a .

A non-empty subset $T \subseteq A^{<\omega}$ is called a *tree* if $s \in T$ and $t \subseteq s$ implies $t \in T$. For each $s \in T$, we write

$$\text{Succ}(s) = \{a \in A : s \frown a \in T\}.$$

We assume throughout that $\text{Succ}(s) \neq \emptyset$ for every $s \in T$. An infinite sequence $x = (a_n)_{n=0}^{\infty} \in A^{\omega}$ is a *play* in T if $(a_0, \dots, a_{n-1}) \in T$ for all $n \geq 1$. The set of all such plays is denoted by $[T]$.

Definition 1.2. An *infinite game* is a pair (T, W) , where $T \subseteq A^{<\omega}$ is a tree and $W \subseteq [T]$. Two players, I and II, alternately choose elements $a_n \in A$ such that $(a_0, \dots, a_n) \in T$. Player I chooses a_{2n} ; Player II chooses a_{2n+1} ($n = 0, 1, 2, \dots$). Player II *wins* the resulting play $x = (a_n) \in [T]$ if $x \in W$; otherwise, Player I wins. The elements of T are called *positions*.

A *strategy* for Player I is a function σ defined on the positions of even length such that $\sigma(s) \in \text{Succ}(s)$. A play or position is *consistent* with σ if $a_{2n} = \sigma(a_0, \dots, a_{2n-1})$ whenever a_{2n} is defined. We say σ is a *winning strategy* for Player I if every play consistent with σ lies in $[T] \setminus W$.

A strategy τ for Player II and the notion of a winning strategy for Player II are defined analogously.



Remarks 1.3. (1) In these notes W is the set of plays won by Player II. Many texts, including [Kec95], use the opposite convention, in which the payoff set is Player I's. Ours suits the games below, where Player II is the player who wants a non-empty intersection.

- (2) The two players need not choose from the same kind of object: if Player I chooses from a set A_I and Player II from a set A_{II} , take $A = A_I \cup A_{II}$ and let the tree T record which moves are legal.

- (3) At most one player has a winning strategy: if σ and τ were winning strategies for Players I and II, the unique play consistent with both would lie both outside and inside W .

2. THE BANACH–MAZUR GAME

Definition 2.1. Let X be a topological space, and let $\mathcal{O}^+(X)$ be the collection of all non-empty open subsets of X . The *Banach–Mazur game* on X , denoted by $\text{BM}(X)$, is the game (T, W) where:

- (1) $A = \mathcal{O}^+(X)$.
- (2) T consists of all finite sequences $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \dots$ of sets in $\mathcal{O}^+(X)$.
- (3) $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^{\infty} U_n \neq \emptyset\}$.

Thus Player I plays the sets U_n , Player II plays the sets V_n , and Player II wins if the sets played have a point in common.

Example 2.2. (1) In $\text{BM}(\mathbb{Q})$, Player I has a winning strategy. Fix an enumeration $q_0, q_1, \dots$ of $\mathbb{Q}$, and let Player I play $U_0 = \mathbb{Q} \setminus \{q_0\}$ and $U_n = V_{n-1} \setminus \{q_n\}$ for $n \geq 1$. These sets are open and non-empty, and every q_n is missing from U_n , so $\bigcap_n U_n = \emptyset$.
 (2) In $\text{BM}(\mathbb{R})$, Player II has a winning strategy: in answer to U_n , she plays a bounded open interval V_n with $\overline{V_n} \subseteq U_n$. The compact sets $\overline{V_n}$ are nested, so these sets have a common point, and that point lies in every U_n .

Theorem 2.3. *Let X be a topological space.*

- (1) *Player I has a winning strategy in $\text{BM}(X)$ if and only if X contains a non-empty open set of the first category.*
- (2) *X is a Baire space if and only if Player I has no winning strategy in $\text{BM}(X)$.*

Proof. By Theorem 0.3, part (2) is an immediate consequence of part (1).

Suppose first that $G \subseteq X$ is open, non-empty, and of the first category. Then $G = \bigcup_{n=0}^{\infty} E_n$, where each E_n is nowhere dense. Player I plays $U_0 = G$. If Player II responds with $V_0 \subseteq U_0$, the set $V_0 \setminus \overline{E_0}$ is open and non-empty. Player I may therefore choose a non-empty open set $U_1 \subseteq V_0 \setminus \overline{E_0}$. In general, given V_{n-1} , Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \quad U_n \neq \emptyset.$$

This defines a strategy σ for Player I. For any play consistent with σ , we have $U_n \cap E_{n-1} = \emptyset$ for all $n \geq 1$, so that

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus σ is a winning strategy for Player I.

Conversely, suppose Player I has a winning strategy σ , and put $U_0 = \sigma(\emptyset)$. We show that U_0 is of the first category. For a position p of odd length $2n+1$ consistent with σ , write U^p for the last entry of p , the set U_n played by Player I.

We build, for each $n \geq 0$, a family $\mathcal{P}_n$ of positions of length $2n+1$ consistent with σ , such that:

- (a) the sets U^p , $p \in \mathcal{P}_n$, are pairwise disjoint;
- (b) their union $D_n = \bigcup_{p \in \mathcal{P}_n} U^p$ is dense in U_0 ;
- (c) every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$.

Let $\mathcal{P}_0 = \{(U_0)\}$. Given $\mathcal{P}_n$, fix $q \in \mathcal{P}_n$. Each non-empty open $V \subseteq U^q$ is a legal move for Player II at q , and leads to the position $q \frown V \frown \sigma(q \frown V)$ of length $2n + 3$. By Zorn's lemma, choose a family $\mathcal{P}(q)$ of such positions, maximal with respect to the property that their last entries are pairwise disjoint. The union of these last entries is dense in U^q : otherwise some non-empty open $W \subseteq U^q$ misses all of them, and then the position $q \frown W \frown \sigma(q \frown W)$, whose last entry lies in W , could be added to $\mathcal{P}(q)$, contradicting maximality. Put $\mathcal{P}_{n+1} = \bigcup_{q \in \mathcal{P}_n} \mathcal{P}(q)$. Property (c) holds by construction. Property (a) holds because last entries of positions in the same $\mathcal{P}(q)$ are disjoint, and those in $\mathcal{P}(q)$ and $\mathcal{P}(q')$ with $q \neq q'$ lie in the disjoint sets U^q and $U^{q'}$. For (b), argue by induction on n , the case $n = 0$ being trivial. A non-empty open $W \subseteq U_0$ meets D_n , hence $W \cap U^q \neq \emptyset$ for some $q \in \mathcal{P}_n$. The union of the last entries of the positions in $\mathcal{P}(q)$ is dense in U^q , so that union meets $W \cap U^q$; since that union is contained in D_{n+1} , the set W meets D_{n+1} .

Each D_n is open and dense in U_0 , so $U_0 \setminus D_n$ is nowhere dense. Suppose U_0 were of the second category. Then U_0 is not contained in the first-category set $\bigcup_n (U_0 \setminus D_n)$, so there is a point $x \in \bigcap_n D_n$. By (a), for each n there is exactly one $p_n \in \mathcal{P}_n$ with $x \in U^{p_n}$. By (c), p_{n+1} extends some $q \in \mathcal{P}_n$, and $x \in U^{p_{n+1}} \subseteq U^q$, so $q = p_n$ by (a). Thus the positions $p_0 \subseteq p_1 \subseteq \dots$ fit together into a single play consistent with σ , and $x \in \bigcap_n U^{p_n}$ lies in every set played by Player I. This contradicts the assumption that σ is winning. Hence U_0 is a non-empty open set of the first category, and the proof is complete. $\textcolor{teal}{\smileyemote}$



Remark 2.4. Part (2) does not say that Player II has a winning strategy in $\text{BM}(X)$ when X is a Baire space; in general neither player need have one (see [Tel87]).

3. THE STRONG CHOQUET GAME

Definition 3.1. Let X be a topological space. The *strong Choquet game* on X , denoted by $\text{Ch}(X)$, is played as follows. Player I chooses a pair (x_0, U_0) with $U_0 \in \mathcal{O}^+(X)$ and $x_0 \in U_0$. Player II responds with $V_0 \in \mathcal{O}^+(X)$ such that $x_0 \in V_0 \subseteq U_0$. At step $n \geq 1$, Player I chooses (x_n, U_n) such that $x_n \in U_n \subseteq V_{n-1}$, and Player II responds with $V_n \in \mathcal{O}^+(X)$ such that $x_n \in V_n \subseteq U_n$. Player II wins the play if $\bigcap_{n=0}^\infty U_n \neq \emptyset$.

In the format of Definition 1.2, Player I chooses from $X \times \mathcal{O}^+(X)$ and Player II from $\mathcal{O}^+(X)$. If the points x_n are omitted, what remains is exactly the Banach–Mazur game $\text{BM}(X)$, which in this context is also called the *Choquet game*. Following Choquet, who called the two players β (the first) and α (the second), X is called *α -favorable* if Player II has a winning strategy in $\text{BM}(X)$, and *strongly α -favorable* if Player II has a winning strategy in $\text{Ch}(X)$; Kechris [Kec95] calls such spaces *Choquet* and *strong Choquet* spaces.

Lemma 3.2. *If X is strongly α -favorable, then X is α -favorable.*

Proof. Let τ be a winning strategy for Player II in $\text{Ch}(X)$. In $\text{BM}(X)$, when Player I plays U_n , Player II picks any point $x_n \in U_n$ and answers with the set that τ gives in reply to (x_n, U_n) . The moves (x_n, U_n) are legal in $\text{Ch}(X)$, because $x_n \in U_n \subseteq V_{n-1}$, so the sets U_n played are those of a play of $\text{Ch}(X)$ consistent with τ , and their intersection is non-empty. $\textcolor{teal}{\smileyemote}$

Example 3.3. Player I wins $\text{Ch}(\mathbb{Q})$: he follows the strategy of Example 2.2(1), adding any point of each set he plays.

Theorem 3.4 (G. Choquet). *Let X be a metrizable space. Then X is completely metrizable if and only if X is strongly α -favorable.*

Proof. Suppose first that X is completely metrizable, and fix a compatible complete metric d on X . We define a strategy τ for Player II. When Player I plays (x_n, U_n) , Player II chooses r_n with $0 < r_n < 2^{-n}$ such that the closed ball $\overline{B}(x_n, r_n)$ is contained in U_n , and plays $V_n = B(x_n, r_n)$.

Consider a play consistent with τ . For $m > n$ we have $x_m \in U_m \subseteq V_n = B(x_n, r_n)$, so $d(x_m, x_n) < 2^{-n}$. Thus (x_n) is a Cauchy sequence, converging to some $x \in X$. For each n , the points x_m with $m > n$ lie in $\overline{B}(x_{n+1}, r_{n+1})$ (since $x_m \in V_{n+1}$ for $m > n + 1$), so $x \in \overline{B}(x_{n+1}, r_{n+1}) \subseteq U_{n+1} \subseteq U_n$. Hence $x \in \bigcap_n U_n$, and τ is a winning strategy for Player II.

For the converse we only indicate the structure of the argument; full details, for separable X , are in [Kec95, Section 8]. For non-separable X the separable argument does not transfer verbatim: arranging the positions at each level so that they cover X and still fit together into plays requires the paracompactness of metric spaces (A. H. Stone's theorem that every open cover of a metric space has a σ -discrete open refinement). Let $\tilde{X}$ be the completion of X with respect to a compatible metric. By Alexandrov's theorem (1924), every G_δ subset of a complete metric space is completely metrizable, so we need only show that X is a G_δ subset of $\tilde{X}$. Given a winning strategy τ for Player II, one may assume that the sets V_n produced by τ have diameter at most 2^{-n} (Player II can always shrink her move to a small ball around x_n). Each open set $V \subseteq X$ is the trace on X of an open set $\tilde{V} \subseteq \tilde{X}$ of the same diameter, and one uses τ to select, level by level, positions whose sets $\tilde{V}$ form open sets $W_n \supseteq X$ with the following property: a point of $\tilde{X}$ lying in every W_n is captured by a single play consistent with τ . Since τ is winning and the diameters tend to 0, such a point is the common point of that play, and so lies in X ; thus $X = \bigcap_n W_n$. The points x_n are what make $W_n \supseteq X$ possible: whatever point of X one wants to capture, Player I can play that point. ☺



Remark 3.5. The points x_n cannot be dropped: an α -favorable metrizable space need not be completely metrizable. Let $X = \{(a, b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$. The open upper half plane H is dense in X , and Player II wins $\text{BM}(X)$ by always answering with an open disk whose closure lies in $U_n \cap H$ and whose radius is less than 2^{-n} . But X is not completely metrizable, since the closed subspace $\mathbb{Q} \times \{0\}$ of X is not even a Baire space. By Theorem 3.4, Player II has no winning strategy in $\text{Ch}(X)$: there, Player I can choose his points on the axis, and Player II must answer with sets containing them.

Corollary 3.6 (Baire category theorem). *Every completely metrizable space is a Baire space.*

Proof. By Theorem 3.4 and Lemma 3.2, Player II has a winning strategy in $\text{BM}(X)$. Then Player I has none, and X is a Baire space by Theorem 2.3. ☺

APPENDIX A. HISTORICAL REMARKS

Baire category. Theorem 0.3 is an elementary reformulation of the definition and carries no attribution. Theorem 0.3 should not be confused with the *Banach category theorem* [Ban30], which is deeper: in any topological space, the union of an arbitrary family of open sets of the first category is again of the first category. The title of Oxtoby's paper [Oxt57] refers to this theorem.

The Banach–Mazur and strong Choquet games. The Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81]: two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets a given set E . Mazur observed that the first player wins if E is comeager in some interval and

that the second player wins if E is of the first category, and conjectured the converses. Banach recorded on 4 August 1935 that the conjecture is true, but his proof was never published. The first published proof is Oxtoby's [Oxt57]. He showed that in an arbitrary topological space the second player has a winning strategy exactly when E is of the first category; the other converse, that the first player wins exactly when E is comeager in some non-empty open set, requires further hypotheses on the space, and holds for instance in complete metric spaces. In the game $\text{BM}(X)$ of these notes the roles are arranged so that the target set is X itself: Player I plays the part of Mazur's second player on the open set U_0 he chooses first, and Theorem 2.3 is Oxtoby's theorem applied locally. See [Tel87] for a survey.

The strong Choquet game goes back to Choquet's work on Baire spaces [Cho58], and is expounded, together with Theorem 3.4, in his *Lectures on Analysis* [Cho69], where the players are called β and α . Kechris states Theorem 3.4 for separable metrizable spaces, where the theorem characterizes the Polish spaces [Kec95, Section 8]; the version given here, for arbitrary metrizable spaces, holds as well.

The notation $[\mathbb{N}]^\omega$ and the partition calculus. The companion notes on Ramsey theory play the Banach–Mazur game of Section 2 on the Ellentuck space of infinite subsets of $\mathbb{N}$, whose notation differs from that of Section 1; this paragraph explains the notation. For a set S and a cardinal κ , one writes $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$ and $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$; thus $[\mathbb{N}]^\omega$ and $[\mathbb{N}]^{<\omega}$ are the families of infinite and of finite subsets of $\mathbb{N}$, as opposed to the sequence spaces A^ω and $A^{<\omega}$ of Section 1. This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which $\kappa \rightarrow (\lambda)_\mu^r$ means: for $|S| = \kappa$, every partition of $[S]^r$ into μ pieces has a set $H \subseteq S$ with $|H| = \lambda$ and $[H]^r$ inside a single piece. Ramsey's theorem becomes $\omega \rightarrow (\omega)_k^r$ for finite r and k . For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned $[\mathbb{N}]^\omega$ into two pieces with no infinite homogeneous set [ER52]. Positive results recover $\omega \rightarrow (\omega)_2^\omega$ for definable partitions: Galvin and Prikry for Borel partitions [GP73], Silver for analytic ones [Sil70], and Ellentuck for partitions with the Baire property in the Ellentuck topology [Ell74]; in the companion notes on Ramsey theory, the Banach–Mazur game on the Ellentuck space is the bridge to these theorems. The notation, and the school of infinite combinatorics that Erdős and his collaborators built around the notation, is set out in the monograph of Erdős, Hajnal, Máté, and Rado [EHMR84].

REFERENCES

- [Ban30] S. Banach, *Théorème sur les ensembles de première catégorie*, Fund. Math. **16** (1930), 395–398.
- [Cho58] G. Choquet, *Une classe régulière d'espaces de Baire*, C. R. Acad. Sci. Paris **246** (1958), 218–220.
- [Cho69] ———, *Lectures on analysis. Vol. I: Integration and topological vector spaces*, Mathematics Lecture Note Series, W. A. Benjamin, Inc., New York–Amsterdam, 1969.
- [EHMR84] P. Erdős, A. Hajnal, A. Máté, and R. Rado, *Combinatorial set theory: Partition relations for cardinals*, Studies in Logic and the Foundations of Mathematics, vol. 106, North-Holland Publishing Co., Amsterdam, 1984.
- [Ell74] E. Ellentuck, *A new proof that analytic sets are Ramsey*, J. Symbolic Logic **39** (1974), no. 1, 163–165.
- [ER52] P. Erdős and R. Rado, *Combinatorial theorems on classifications of subsets of a given set*, Proc. London Math. Soc. (3) **2** (1952), 417–439.
- [ER56] ———, *A partition calculus in set theory*, Bull. Amer. Math. Soc. **62** (1956), no. 5, 427–489.
- [GP73] F. Galvin and K. Prikry, *Borel sets and Ramsey's theorem*, J. Symbolic Logic **38** (1973), no. 2, 193–198.
- [Kec95] A. S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.
- [Mau81] R. D. Mauldin (ed.), *The Scottish Book: Mathematics from the Scottish Café*, Birkhäuser, Boston, Mass., 1981.

- [Oxt57] J. C. Oxtoby, *The Banach–Mazur game and Banach category theorem*, Contributions to the Theory of Games, vol. III, Annals of Mathematics Studies, no. 39, Princeton University Press, Princeton, N.J., 1957, pp. 159–163.
- [Oxt80] ———, *Measure and category*, second ed., Graduate Texts in Mathematics, vol. 2, Springer-Verlag, New York, 1980.
- [Sil70] J. Silver, *Every analytic set is Ramsey*, J. Symbolic Logic **35** (1970), no. 1, 60–64.
- [Tel87] R. Telgársky, *Topological games: On the 50th anniversary of the Banach–Mazur game*, Rocky Mountain J. Math. **17** (1987), no. 2, 227–276.